Reg No……..

Name……..

**Mahatma Gandhi University Kottayam**

**Third Semester MGU – UGP (HONOURS) Regular Examination ……2025**

**MG3DSCECO201 – Quantitative Economics- I**

**2024 Admission Onwards**

Duration: 2 hours Maximum marks: 70

**(Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I) Appreciation (Ap) and Skill (S))**

*Students should attempt at least one question from each course outcome to enhance their overall outcome availability.*

*[Learning Domain] [Cos No (s)]*

**Part A**

**Multiple Choice Questions**

**Answer all questions**

**Each question carries 1 mark**

1. Which of the following is a **linear equation**? [U][1]
   a) $2x+3=0$
   b) $x^2-4=0$
   c) $y=e^x$
   d) $\log x = 2$
2. The value of $(x^3)^2$ is: [U][1]
   a) $x^5$
   b) $x^6$
   c) $x^9$
   d) $x^8$

3. If $\log_{10}$ =x, then x=? [U][1]

   a) 1

   b) 2

   c) 10

   d) 100

4. If A={a,b,c}, the number of subsets of A is: [U][1]

   a) 3

   b) 6

   c) 7

   d) 8

5. A die is thrown once. The probability of getting an even number is: [A][2]

   a) 1/6

   b) 1/2

   c) 1/3

   d. 2/3

6. A matrix with only one row is called: [U][2]

   a) Column matrix

   b) Row matrix

   c) Square matrix

   d) Null matrix

7. If A=$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ then det A= ? [U][2]

   a) −2-

   b) −5

   c) 2

   d) 5

8. The demand and supply equations for a good are: [C][5]

$$Qd = 30 - P; Qs = 2P$$

   Find the equilibrium price.

   A) 6
   B) 8
   C) 10
   D) 12

9. Derivative of a constant is always: . [U][2]

   a) Constant

   b) 1

   c) 0

   d) Undefined

10. If the total cost function is C=5q+100 , the marginal cost (MC) is: [E][3]

    a) 5q

    b) q+100

    c) 5

    d) 100

**Part B**

**Short Answer Questions (50 words)**

**Answer any ten questions**

**Each question carries 3 marks**

11. How to solve quadratic equations? [U][1]
12. Distinguish between super set and subset. [U][1]
13. What are the different types of relations? [U][1]
14. If set A = {1, 3, 5, 7, 9} and set B = {1, 2, 3, 4, 5}, find A ∪ B and A ∩ B. [U][1
15. Sets J = {2, 4, 6, 8, 10} and K = {3, 6, 9, 12}. Find the Cartesian product J × K. [U][1]
16. If $A = \begin{bmatrix} 7 & 8 & 9 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ $B = \begin{bmatrix} 6 & 5 & 4 \\ 3 & 2 & 1 \\ 9 & 8 & 7 \end{bmatrix}$ Find A+B and A-B [U][2]
17. From a pack of 52 playing cards one is drawn at a random. Find the probability that it is (a) a black card (b) a diamond card. [U][2]
18. Define random experiment, sample space and event [U][2]
19. Using the following sets A={1,3,5}, B={2,4,6}, and U={1,2,3,4,5,6}, find the complements of [U][1]
    **a.** A′
    **b.** B′
20. Define equally likely and exhaustive events. [U][2]

21. Write any three properties of determinants. [U][2]

22. Distinguish between symmetric and skew symmetric matrix with examples. [U][1]

23. Distinguish between maxima and minima. [U][2]

24. Find the limits for the following functions: [A][4]

    a. $\lim_{x \to 1}[X^3(X+4)]$

    b. $\lim_{x \to 4} \frac{3X^2 - 5X}{X+6}$

    c. $\lim_{x \to 2} \sqrt{6X^3 + 1}$

25. If the total cost function is C=200+5q+$q^2$, find the marginal cost (MC). [E][3]

## Part C

## Short Essay Type Questions (160 words)

## Answer any 5 questions

## Each question carries 6 marks

26. Maximize profits π for the firm, given total revenue $R = 4000Q - 33Q^2$ and total cost $C = 2Q^3 - 3Q^2 + 400Q + 5000$, assuming Q>0. [E][3]

27. Consider the sets: [U][1]

    P={1,2,3,4}

    Q={3,4,5,6}

    R={4,5,6,7}

Find the following:

    a. P∪(Q∩R)

    b. (P∪Q)∩(P∪R)

28. Solve the equation {U][1]

$$5X^2 + 23X + 12 = 0$$

29. Find the equilibrium price and equilibrium quantity [C][5]

$$Qs = -20 + 3p$$

$$Qd = 220 - 5p$$

30. A box contains 6 red balls, 7 white balls and 3 green balls. If a ball is drawn at random, what is the probability that it is red or green. [U][2]

31. Five men in a company of 20 graduates. If 3 of them are selected at random what is the probability that (a) all of them are graduates (b) 2 of them are graduates? [U][2]

32. Find AB and BA of the matrix. [U][2]

A = $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ $\begin{bmatrix} -7 & -8 \\ 9 & 10 \\ 0 & 11 \end{bmatrix}$

33. Solve the equations using Cramer's Rule [U][5]

$$x - 3Y + 2Z = 8$$

$$3X + 4Y + Z = 5$$

$$-4X + 2Y - 9Z = 2$$

34. Find the inverse of the matrix [A][4]

a. $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & 2 & 1 \end{bmatrix}$

35. Find the relative extrema for the following functions by [E][3]

a. Finding the critical values, and

b. Determining if it the critical values the function is at a relative maximum or minimum.

1. $F(x) = -7X^2 + 126X - 23$

2. $Y = 3X^3 - 36X^2 + 135\,X - 13$